UP Special Topic 4 — Economy, Industrial Policy, Mineral Resources, Infrastructure and ODOP

Current Affairs

Year	Fact	Why it matters
Rolling	Refresh edition-bound figures from latest Census / Survey / State CA desks.	Do not invent interim totals.

Teaching cards

2.1 Macroeconomic Profile & Sectoral Structure of Uttar Pradesh

A. State Gross Domestic Product (GSDP) & Economic Ambition

- **Economy Scale**: Uttar Pradesh is the **third largest state economy in India** by nominal GSDP (after Maharashtra and Tamil Nadu / Gujarat), contributing ~8.5%–9.0% to India's national GDP.
- **The 1 Trillion Dollar Target**: The Government of Uttar Pradesh has formally set a strategic goal to scale the state's GSDP to **1 Trillion USD**, anchoring development around: 1. Expressways and multimodal logistics corridors. 2. The UP Defence Industrial Corridor and dedicated manufacturing clusters. 3. Export promotion via the ODOP scheme. 4. World-class civil aviation infrastructure (Noida International Airport at Jewar, Ayodhya, Varanasi).
- **Sectoral Contribution**:
- **Primary Sector (Agriculture & Allied)**: Contributes ~24%–26% of GSDP, employing nearly 45%–50% of the state's total workforce.
- **Secondary Sector (Manufacturing, Mining, Construction, Electricity)**: Contributes ~24%–26% of GSDP.
- **Tertiary Sector (Services, IT, Banking, Tourism, Real Estate)**: Contributes the largest share (~48%–50%) of GSDP.
- **Per Capita Income Dynamics**: Despite high aggregate GSDP, UP's per capita Net State Domestic Product (NSDP) remains below the national average due to the massive population base (~24 crore).

B. Regional Economic Disparities

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│                    REGIONAL ECONOMIC PROFILE OF UP               │
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│ Region          │ Economic Characteristics & Key Engines         │
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│ 1. Western UP   │ Highly developed; Green Revolution agriculture (Wheat/  │
│                 │ Cane); dense MSME hubs; NCR industrial belt (Noida/     │
│                 │ Ghaziabad/Meerut); highest per capita income in UP.     │
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2. Central UP	Moderate development; Kanpur industrial node (Leather/ Textiles); Lucknow administrative, IT and medical hub; fertile agricultural plains (Awadh).
3. Purvanchal (Eastern UP)	Densely populated; agrarian economy with small land holdings; historical out-migration of labour; massive infrastructure push (Purvanchal Exp, Gorakhpur AIIMS/ Fertilizer Plant, Varanasi Multimodal Freight Terminal).
4. Bundelkhand	Semi-arid plateau; drought-prone; agrarian economy based on pulses/oilseeds; low urbanization; Defence Corridor nodes (Jhansi/Chitrakoot) and solar parks leading push.

2.2 Major Traditional & Modern Industries

A. The Agro-Based Sugar Industry

- **Pioneering Legacy**: India's first modern vacuum-pan sugar mill was established in **1903 at Pratappur in Deoria district** of Uttar Pradesh.
- **Production Dynamics**: UP is the **largest sugarcane-producing state in India** (~45%–50% of national production) and alternates with Maharashtra for the #1 spot in refined sugar output.
- *Distinction*: UP leads in **total acreage and sugarcane volume** (sub-tropical cane belt), whereas Maharashtra leads in **cane yield per hectare and recovery rate/cooperative structure** (tropical cane belt).
- **Ethanol Blending Programme**: UP is India's leading producer of **molasses-based fuel ethanol**, contributing massively to the National Biofuel Policy target of 20% ethanol blending in petrol.

B. Leather, Textiles & Cottage Clusters

1. **Leather & Saddlery**: - Centred at **Kanpur, Agra, and Unnao**. - Kanpur produces world-class equestrian saddlery and safety boots; Agra specializes in commercial footwear manufacturing for domestic and export retail.
2. **Textiles & Handlooms**: - Kanpur was historically dubbed the **"Manchester of the East"** due to British textile mills along the Ganga. - Handloom and powerloom weaving clusters flourish in **Varanasi (Banarasi Silk)**, **Mau**, **Azamgarh (Mubarakpur)**, **Tanda (Ambedkar Nagar)**, and **Meerut**.
3. **Glassware at Firozabad**: - Known as *Suhag Nagari*, producing millions of colourful glass bangles, blown glassware, and scientific glass instruments.
4. **Carpets at Bhadohi-Mirzapur**: - "Carpet City" Bhadohi houses the Indian Institute of Carpet Technology (IICT). The region accounts for the majority of India's hand-knotted export carpets.

2.3 Mineral Wealth and Geological Distribution

Uttar Pradesh is generally poor in metallic minerals as over 70% of its area is covered by deep quaternary Gangetic alluvium. Minerals are confined almost exclusively to the **Southern Plateau and Hill Region** (Vindhyan and Precambrian formations):

1. **Coal**: Confined to the **Singrauli coalfields** in the southern part of **Sonbhadra district**.
2. **Limestone**: Massive deposits in the Vindhyan rock formations at **Kajrahat and Rohtas (Sonbhadra)** and **Bhalua, Guruma-Kanach (Mirzapur)**; supplies the regional cement industry.
3. **Silica Sand (Glass Sand)**: Extensively quarried from the weathered quartzite sandstones of the Vindhyan range at **Shankargarh, Lohgara (Prayagraj)** and **Banda/Chitrakoot**.
4. **Bauxite (Aluminium Ore)**: Found in **Banda, Chitrakoot, and Varanasi (Chandauli)**.
5. **Dolomite**: Found in **Sonbhadra (Bari village)** and **Banda**; used as a refractory flux in the steel industry.
6. **Uranium**: Traces discovered in **Lalitpur district**.
7. **Copper**: Found in the **Sonrai region of Lalitpur district**.
8. **Diaspore and Pyrophyllite**: Extensive reserves in **Jhansi, Lalitpur, Mahoba, and Hamirpur**; UP is India's leading producer.
9. **Rock Phosphate**: Extracted in **Lalitpur district** (used in phosphatic fertilizers).
10. **Gold Traces**: Found in the river sands of **Sharda and Ramganga**, and mineralized quartz veins in **Son Pahadi / Hardi area (Sonbhadra)** and **Lalitpur**.

2.4 Energy Profile: Thermal, Nuclear, Hydro & Renewable

A. Thermal Power Infrastructure

- Sonbhadra district is known as the **"Energy Capital of India"** because of the dense concentration of pithead thermal power stations:
- **NTPC Singrauli Super Thermal Power Station (Shaktinagar, Sonbhadra)**: NTPC's first power plant (commissioned 1982).
- **NTPC Rihand Super Thermal Power Station (Rihand Nagar, Sonbhadra)**.
- **Obra Thermal Power Station (Sonbhadra - UPRVUNL)**.
- **Anpara Thermal Power Station (Sonbhadra - UPRVUNL)**: Major state-owned base load station.
- **Harduaganj Thermal Power Station (Aligarh)**: The oldest thermal power station in Uttar Pradesh, established in **1942**.

B. Nuclear Power

- **Narora Atomic Power Station (NAPS)** located at Narora in **Bulandshahr district**.
- Operated by the Nuclear Power Corporation of India Limited (NPCIL).
- Consists of two indigenous Pressurized Heavy Water Reactors (PHWR) of 220 ext{ MWe} each. Uses water from the Lower Ganga Canal for secondary cooling.

C. Renewable Solar Power

- Uttar Pradesh Solar Energy Policy targets significant capacity expansion.
- Large-scale solar parks set up in **Bundelkhand (Jalaun, Jhansi, Lalitpur, Banda)** and **Mirzapur** (the 75 MW Dadar Kalan solar power plant was inaugurated jointly with French assistance).

2.5 Strategic Industrial Corridors & Flagship Policies

A. Uttar Pradesh Defence Industrial Corridor (UPDIC)

- Announced in the Union Budget 2018–19 to reduce import dependence in aerospace and defence manufacturing.
- Implemented by the **UP Expressways Industrial Development Authority (UPEIDA)** across **6 strategic nodes**: 1. **Lucknow Node**: **BrahMos Aerospace** integration unit for BrahMos-NG cruise missiles and aerospace components. 2. **Kanpur Node**: Small arms, artillery components, defence textiles, and UAV systems. 3. **Jhansi Node**: Bharat Dynamics Limited (BDL) propulsion plant, weapon systems, and tank overhauling. 4. **Aligarh Node**: Precision hardware, locks, drones, and small arm components. 5. **Agra Node**: Aviation parts, logistics, and electronics. 6. **Chitrakoot Node**: Defence testing infrastructure, propellants, and auxiliary units.

B. National Smart Cities Mission in Uttar Pradesh

Under the Central Government's Smart Cities Mission, **10 cities** of Uttar Pradesh were selected: 1. Lucknow 2. Varanasi 3. Prayagraj 4. Agra 5. Kanpur 6. Aligarh 7. Jhansi 8. Moradabad 9. Bareilly 10. Saharanpur *(Note: Ghaziabad, Meerut, Ayodhya, Gorakhpur, Firozabad, Shahjahanpur, and Mathura-Vrindavan were subsequently funded under the State Smart City Scheme, NOT the original Central 10 list).*

Bilingual Terminology -- UP Economy

English	Hindi	Key Anchor
GSDP	राज्य सकल घरेलू उत्पाद	Gross State Domestic Product -- UP ~2nd largest economy
MSMEs	सूक्ष्म, लघु और मध्यम उद्यम	Micro, Small and Medium Enterprises
MSME Cluster	MSME क्लस्टर	Agglomeration of similar industries (Moradabad brass, Varanasi silk)
Moradabad Brass Work	मुरादाबाद का पीतल उद्योग	Largest brass ware export cluster in India
Varanasi Silk	वाराणसी का रेशमी वस्त्र	Banarasi silk sarees; GI tag
Firozabad Glass	फिरोजाबाद का कांच उद्योग	Bangles and glassware capital of India
Agra Leather	आगरा का चमड़ा उद्योग	Major footwear export hub (Agra, Kanpur)
Lucknow Chikankari	लखनऊ की चिकनकारी	GI-tagged delicate embroidery work
Aligarh Lock Industry	अलीगढ़ ताला उद्योग	World's largest lock manufacturing cluster
Gorakhpur Terracotta	गोरखपुर की टेराकोटा	Traditional fired-clay craft; Gorakhpur district
UP ODOP	एक जिला एक उत्पाद	One District One Product scheme (launched 2018)

English	Hindi	Key Anchor
UP Industrial Policy	उत्तर प्रदेश औद्योगिक नीति	Ease of doing business reforms; investment summits
Noida	नोयडा	New Okhla Industrial Development Authority; IT/Electronics hub
YEIDA	यमुना एक्सप्रेसवे औद्योगिक विकास प्राधिकरण	Yamuna Expressway Industrial Development Authority
Film City (Noida)	नोयडा फिल्म सिटी	Asia's largest proposed integrated media production hub

Extended Theory -- UP's Economic Profile

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UP ECONOMY -- KEY SECTORS (Approx. % GSDP)
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Agriculture + Allied: ~20-22%
├─ Sugarcane (largest producer in India)
├─ Wheat (2nd largest producer)
├─ Potato (largest producer)
└─ Milk production (largest in India)

Industry: ~25-28%
├─ Sugar mills (largest number in India)
├─ Leather goods (Agra, Kanpur)
├─ Electronics/IT (Noida, Lucknow)
└─ Textiles (Varanasi silk, Lucknow chikankari)

Services: ~50-52%
├─ IT services (Noida tech corridor)
├─ Tourism (Agra, Varanasi, Prayagraj)
└─ Banking and trade
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Major Industries and GI Tags of UP

Industry / Product	Location	Key Fact
Brass ware	Moradabad	"Peetal Nagari"; largest export cluster
Glass bangles	Firozabad	"Suhag Nagari"; 80%+ of India's bangles
Leather goods	Agra, Kanpur	Agra = footwear; Kanpur = leather goods
Silk sarees	Varanasi	Banarasi silk = GI tag; UNESCO recognition
Chikankari	Lucknow	GI tag; 30+ types of embroidery stitches
Lock making	Aligarh	50%+ of India's padlocks

Industry / Product	Location	Key Fact
Carpet weaving	Bhadohi, Mirzapur	"Carpet City"; major export
Terracotta	Gorakhpur	Distinctive black-clay figurines
Zardozi	Lucknow, Varanasi	Gold/silver thread embroidery; GI tag
Khurja pottery	Bulandshahr	"Ceramic City"; glazed earthenware

ODOP (One District One Product) -- Selected Examples

District	ODOP Product
Moradabad	Brass / Metal Craft
Varanasi	Banarasi Silk / Handloom
Lucknow	Chikankari
Aligarh	Locks and Hardware
Firozabad	Glass / Bangles
Agra	Leather Footwear
Bhadohi	Carpet
Gorakhpur	Terracotta
Kannauj	Perfume (Ittar)
Bareilly	Zari-Zardozi

One-Liner Revision -- UP Economy

#	Fast Fact
1	UP = 2nd largest State economy by GSDP (after Maharashtra)
2	UP = largest sugarcane-producing State
3	UP = largest sugar mill State (over 100 sugar mills)
4	UP = largest potato-producing State
5	UP = largest milk-producing State
6	UP = 2nd largest wheat-producing State (after MP/Punjab)
7	ODOP launched = **2018** (75 districts -- 75 products)
8	Moradabad = **"Peetal Nagari"** (brass ware exports)
9	Firozabad = **"Suhag Nagari"** (80%+ of India's bangles)

#	Fast Fact
10	Aligarh = **"Lock City"** (50%+ India's padlocks)
11	Kanpur = leather goods hub (second to Agra)
12	Bhadohi/Mirzapur = **"Carpet City"**
13	Noida = IT/Electronics corridor; part of NCR
14	Lucknow = Chikankari (GI tag) + Zardozi embroidery
15	Kannauj = **Ittar (perfume) city** of India